## TOPIC 1 Linear Equations and Inequalities

Name _____ Date _____

**Reminders**
**Words:** at least ≥ · at most ≤ · more than > · fewer than <.
**Solve like an equation:** undo + and −, then × and ÷.
**Flip the sign** when you multiply or divide both sides by a negative number.
**Graph:** ○ for < and >, ● for ≤ and ≥. Shade right for > and ≥, left for < and ≤.

**1.** What is the solution set of $-4x > 12$?

**Tip:** ÷ by a negative → flip the sign

**A.** $x > -3$

**B.** $x < 3$

**C.** $x < -3$

**2.** What is the solution set of $2x + 3 \leq x - 1$?

**Tip:** get [[x]] on one side, like an equation

**F.** $x \leq -4$

**G.** $x \geq -4$

**H.** $x \leq -2$

**3.** Which inequality means the same as $-6 \leq x$?

> **Tip:** switch sides → the sign points the other way

**A.** $x \leq -6$

**B.** $x \geq -6$

**C.** $x > -6$

**4.** What value of $x$ makes $3(x - 2) = 9$ true?

> **Tip:** distribute, then undo + and ×

**F.** 3

**G.** 11/3

**H.** 5